JIZHOU HUANG

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RESEARCH OVERVIEW

My primary research interests lie in computational learning theory, with a focus on the PAC-learnability of conditional models and threshold-gate neural networks. My work also explores the learning of interpretable and fair representations, which is central to building reliable and trustworthy predictive systems.

EDUCATION

Aug. 2019 – Aug. 2025 **Ph.D. in Computer Science**
Washington University in St. Louis, MO, USA
Thesis: Learning From Conditional Data Distributions, advised by Brendan Juba

Aug. 2022 – May 2023 **Visiting Scholar at Simon's Institute**
University of California, Berkeley, CA, USA

Aug. 2016 – Dec. 2018 **M.S. in Electrical Engineering & Computer Science**
Washington University in St. Louis, MO, USA
Project: Learning Low-Energy Threshold Gate Networks, advised by Brendan Juba

Sept. 2011 – May 2015 **B.E. in Automation**
Beijing Jiaotong University, Beijing, CHINA
Project: Boosting Massive Image-Search System using FPGA, advised by Shikui Wei

RESEARCH EXPERIENCE

Aug. 2019 – Aug. 2025 **Washington University in St. Louis**, MO, USA
Research Assistant at *Mckelvey School of Engineering*
Advisor: Brendan Juba, Collaborator: Daniel Hsu
Learning From Conditional Data Distributions

- Studied a family of problems whose objective is to model properties of sub-populations of a distribution, typically by jointly learning a sub-population representation along with the target representation, under PAC-learning semantics.
- Developed a computationally efficient algorithm for *label-agnostic* linear classification conditioned on sub-populations represented by halfspaces under Gaussian distributions, with a provable approximation guarantee.
- Defined a family of *well-behaved* distributions that generalizes Gaussian and many other common distributions. Proved that our conditional classification algorithm may achieve comparable performance guarantees under this broader distribution class, and empirically evaluated the algorithm in PyTorch[1].
- Proved that approximating classification tasks conditioned on general halfspaces within a small additive error is intractable under standard cryptographic hardness assumptions.
- Proposed a *personalized prediction scheme* where a dedicated predictor is learned for each incoming query, rather than a universal predictor to predict all future queries.
- Developed an efficient algorithm for personalized prediction that jointly learns a halfspace containing the query point and a sparse linear classifier with provable performance guarantees on that halfspace under well-behaved distributions. The sparse classifier produced by our algorithm achieved competitive performance compared with more complex models in empirical evaluations[2], while offering superior interpretability due to its simplicity.
- Studied the problem of *auditing* subgroup fairness, which aims to determine whether a decision-making process discriminates against any demographic subpopulation.
- Established that the auditing problem remains computationally intractable when subpopulations are represented by membership functions as simple as halfspaces—even under Gaussian distributions—based on standard cryptographic hardness assumptions.

[2]https://github.com/JizhouHuang/conditional-classification-with-homogeneous-selectors
[2]https://github.com/JizhouHuang/personalized-prediction-with-homogeneous-halfspaces

- Reduced the auditing problem for homogeneous halfspace subgroups to learning homogeneous halfspaces under Gaussian distributions. Developed a auditing algorithm with provable guarantees by leveraging a Polynomial-Time Approximation Scheme for learning homogeneous halfspaces.
- Developed an efficient improper auditing algorithm for subgroups represented by conjunctions, which is able to identify an unfairly treated subgroup defined by some polynomial threshold function.

Oct. 2016 – May 2019 **Washington University in St. Louis**, MO, USA
Student Researcher at *Mckelvey School of Engineering*
Advisor: Brendan Juba, Collaborator: Zihao Deng
Learning Low-Energy Threshold Gate Networks
- Studied the PAC-learnability of sparsely activated (low-energy) threshold-gate networks. Investigated the relationship between the Unique Games Conjecture on expander graphs and such low-energy networks.
- Proposed an SVM-like program that captures different activation patterns of the gates, enabling the application of multiple robust clustering methods. Proved a energy upper bound on the number of variables involved in each constraint, which significantly reduced the program's complexity.
- Showed that the set of possible weights for each feasible activation pattern forms a convex cone over the hinge loss, indicating that low-energy networks with different hinge losses can be distinguished by the angle between their vectorized parameters.
- Formulated the problem as a spherical optimization task that fits within the geodesic optimization framework, allowing strong convexity to be encoded into the loss function.

Dec. 2014 – Mar. 2016 **Beijing Jiaotong University**, Beijing, CHINA
Student Researcher at *Electronic Information Engineering Institute*
Advisor: Shikui Wei
Boosting Massive Image-Search System using FPGA
- Built a *k-Nearest Neighbor* based image-searching system on PC using C#, and an image-searching system of the same algorithm on FPGA using C++.
- Applied optimization instructions to the C++ code using Vivado High-Level Synthesis Software, which significantly improved the throughput of the algorithm on FPGA. Integrated the optimized program into a Zedboard embedded system using Vivado Software and set up multiple properties of the system using C on Xilinx SDK.
- Configured AXI Direct Memory Access via Xilinx SDK to achieve high-speed data exchanging between an on-board SD card and Xilinx Artix-7 FPGA, which solved the problems caused by memory deficiency and slow data transfer.

TEACHING EXPERIENCE

Spring 2024 **Washington University in St. Louis**, MO, USA
Teaching Assistant at *Mckelvey School of Engineering*
CSE 513T Theory of Artificial Intelligence and Machine Learning
- Filled in for multiple lectures.
- Graded students' homework and held weekly TA sessions to answer students' questions.

Fall 2021 **Washington University in St. Louis**, MO, USA
Teaching Assistant at *Mckelvey School of Engineering*
CSE 513T Theory of Artificial Intelligence and Machine Learning
- Filled in for multiple lectures.
- Graded students' homework and held weekly TA sessions to answer students' questions.

WORK EXPERIENCE

June 2025 – Present **Software Development Engineer at Meta**, CA, USA

ESR Modeling – IG Reels Recommendation

- Had major contribution to the development of an advanced data sampling project for the ESR MasterBrew (MB) model, supporting complex feature-based population selection and Local Case-Control (LCC) sampling. Its launch significantly improved IG Reels top-line metrics.
- Led the adaptation of the model-side sampling mechanism to the data-provider side, significantly reducing local CPU usage on the model side.
- Drove the study of the impact of Watch-Time-related signals and the development of novel encoding approaches for "Watch Time" and "Video Length" signals.
- Contributed to the development of a co-trained ESR MB model, which jointly trains an LSR model (teacher) to guide the ESR model (student).

LSR Modeling – FB Feed Recommendation

- Drove the deployment of a novel FB Feed Recommendation (CFR) model, significantly pushed the online metrics towards the goal of the LSR Modeling team.
- Applied a SHAP-based feature selection method on both dense and sparse post features of the CFR model, achieved significant improvement on online metrics.
- Drove the development of a search-based user history distillation (Hard-Search) module as a major component of the CFR model (at pre-processing phase). Participated in the development of an attention-based module (in the CFR model) specialized for the distilled user history sequence. These modules together eventually delivered remarkable online metric gain.
- Leveraged a novel user embedding (ViSTA), which encoded life-long cross-surface features, to significantly boost the performance of FB CFR model.
- Rebuilt FB Story-In-Feed Pass-0 model using a new Python-based environment. Non-trivially moved online metrics and enabled MITA-based (Meta's own hardware) inference.

May 2023 – Aug. 2023 **Machine Learning Engineer Intern at Meta**, CA, USA

Path-Based Trigger Network – MRS AI Recommendation

- Created a trigger net retrieval model to enhance Facebook Reels recommendations, modeling user engagement with items through their previously engaged items.
- Proposed and implemented a novel variant of the trigger net using Gram-Schmidt decomposition, resulting in reduced training loss compared to the baseline model.
- Built a serving module for the trigger net, enabling the serving of trigger net for FB reels recommendations. Evaluated both models and refined them based on real-world user traffic.

May 2022 – Aug. 2022 **Machine Learning Engineer Intern at Meta**, CA, USA

Multi-interest Retrieval – MRS Recommendation Core ML

- Worked on building recommendation models to capture different human interests at retrieval stage for Facebook reels.
- Built and trained a two-tower model consist of multiple sparse sub-module.
- Analyzed the pair-wise correlations between individual sparse module and verified the interests separation across heads.

PUBLICATIONS

- **J. Huang**, D. Hsu, and B. Juba. Distribution-Specific Auditing For Subgroup Fairness. *The 5th annual Symposium on Foundations of Responsible Computing*, Cambridge, MA, USA, 2024.
- **J. Huang**, B. Juba. Distribution-Specific Agnostic Conditional Classification With Halfspaces. *The 13th International Conference on Learning Representations*, Singapore, 2025.
- **J. Huang**, B. Juba. Personalized Prediction By Learning Halfspace Reference Classes Under Well-Behaved Distribution. Preprint, arXiv: https://arxiv.org/abs/2509.15592.
- **J. Huang**, B. Juba. Near-Optimal Cryptographic Hardness of Learning With Homogeneous Halfspaces Under Gaussian Marginals. Preprint, arXiv: https://arxiv.org/abs/2604.26446.

TALKS

- "Distribution-Specific Auditing For Subgroup Fairness" in *The 5th annual Symposium on Foundations of Responsible Computing*, Harvard University, Cambridge, MA, June 13, 2024.
- "Subgroup-based Fairness Auditing - from a computational complexity view" in *Doctoral Student Research Seminar(DSS)*, Washington University in St. Louis, St. Louis, MO, November 17, 2023.
- "An Efficient Algorithm For Fairness Auditing" in *Doctoral Student Research Seminar(DSS)*, Washington University in St. Louis, St. Louis, MO, October 22, 2021.

AWARDS

- FORC 2024 Best Student Paper

WORKSHOPS

- "Auditing Subgroup Fairness on Nice Distributions", *NSF Fairness in AI PI Meeting*, Arlington, VA, USA, 2023.